

Introduction to ARM systems



R. Ferrero

Politecnico di Torino

Dipartimento di Automatica e Informatica (DAUIN)

Torino - Italy

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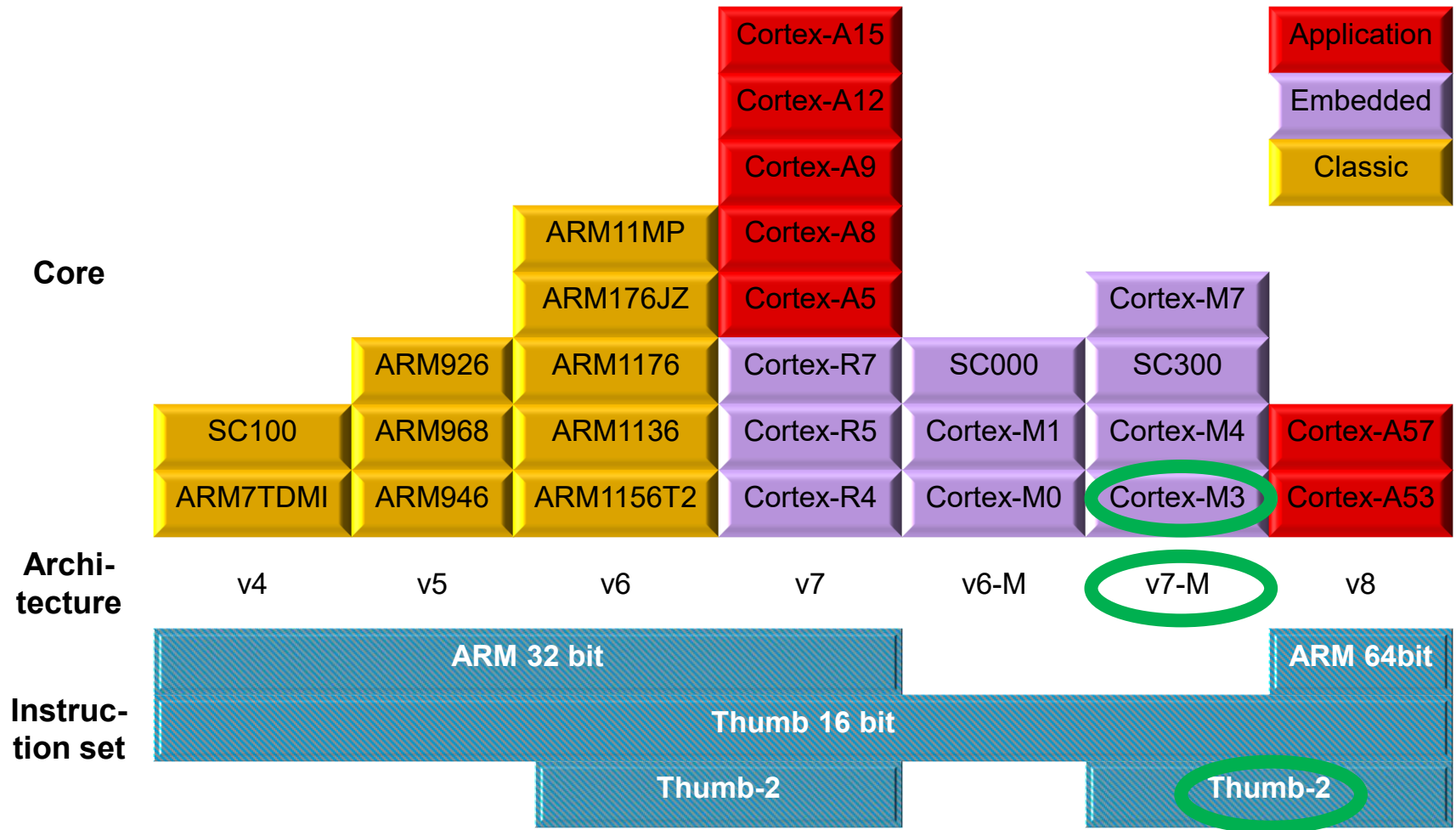
History: Acorn RISC Machines

- 1985: ARM1, first ARM processor (v1)
 - designed by Acorn Computers
 - produced by VLSI Technology (now NXP)
- 1987-1989: ARM2 and ARM3 (v2)
 - used by Acorn in its desktop PCs
 - sold by VLSI as part of processor chip sets
- 1990: Acorn and Apple found ARM Ltd.
 - new business model: manufacturing rights instead of processors
 - VLSI become the first licensee.

History: Advanced RISC Machines

- 1991: ARM6 (v3)
- 1993: ARM7 (v4)
 - ARM7TDMI: Debug and ICE (In-Circuit Emulation)
 - new compressed instruction set: Thumb
- 1996-2006: ARM8 - ARM11 (v5, v6)
- 2004-today: Cortex cores (v7, v8)
 - Cortex-A: high-end applications (smartphones)
 - Cortex-R: real-time and safety-critical applications
 - Cortex-M: microcontrollers (Cortex-M3 LPC1768).

ARM families and architectures



ARM today



Partnership

Silicon Partners



Design Support Partners



Software, Training and Consortia Partners



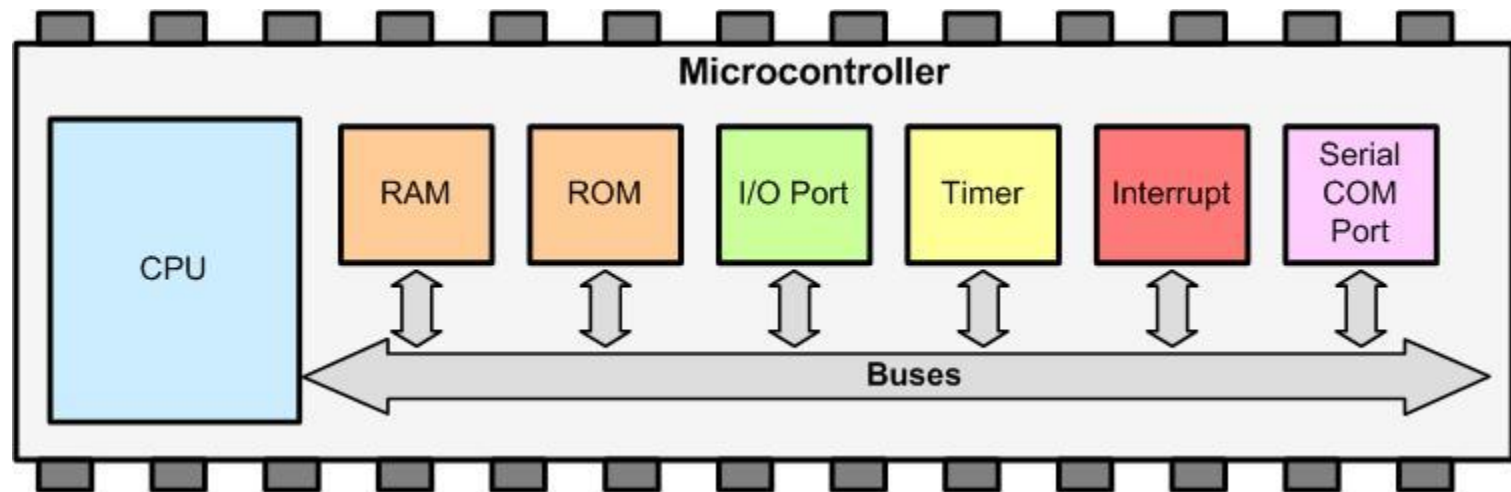
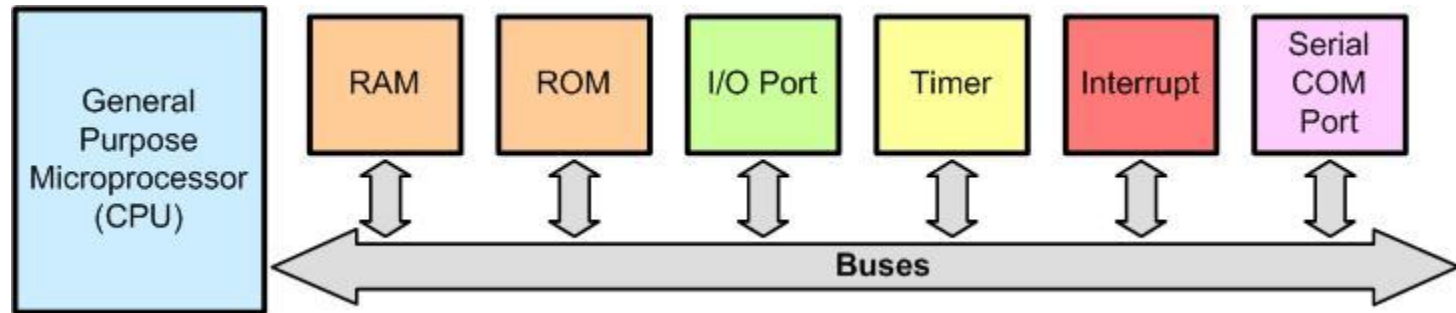
ARM powered products



ARM-based processors

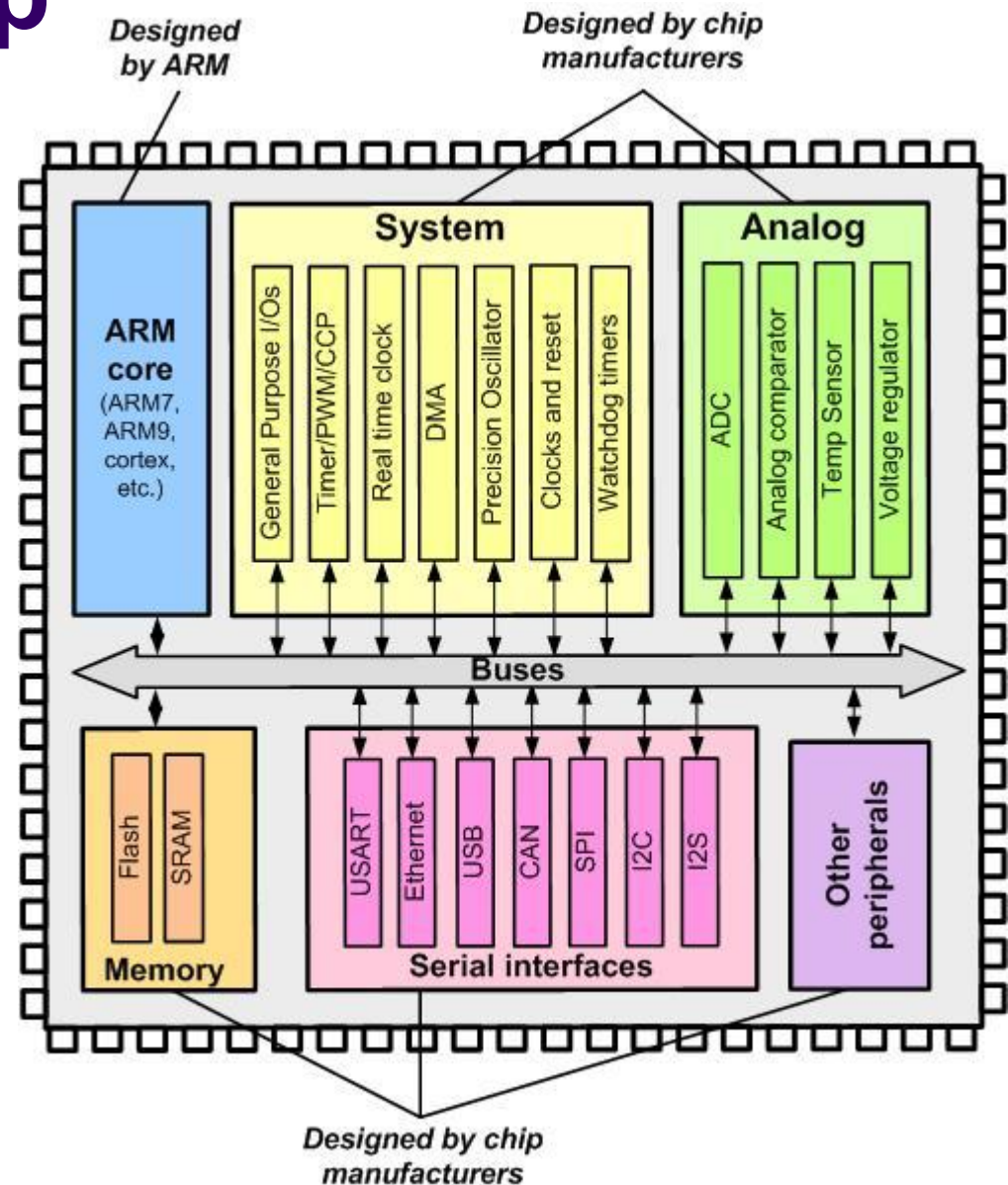
- ARM has seldom delivered stand-alone devices with its own microprocessor.
- ARM mainly sells cores, to be used for integration in microcontrollers or Systems on Chip (SoCs).
 - *Hard cores*: ARM provides a physical layout implemented in a given technology.
 - *Soft cores*: ARM provides a high-level description, which can be then synthesized to any technology by the designer.

Microprocessor Vs. microcontroller

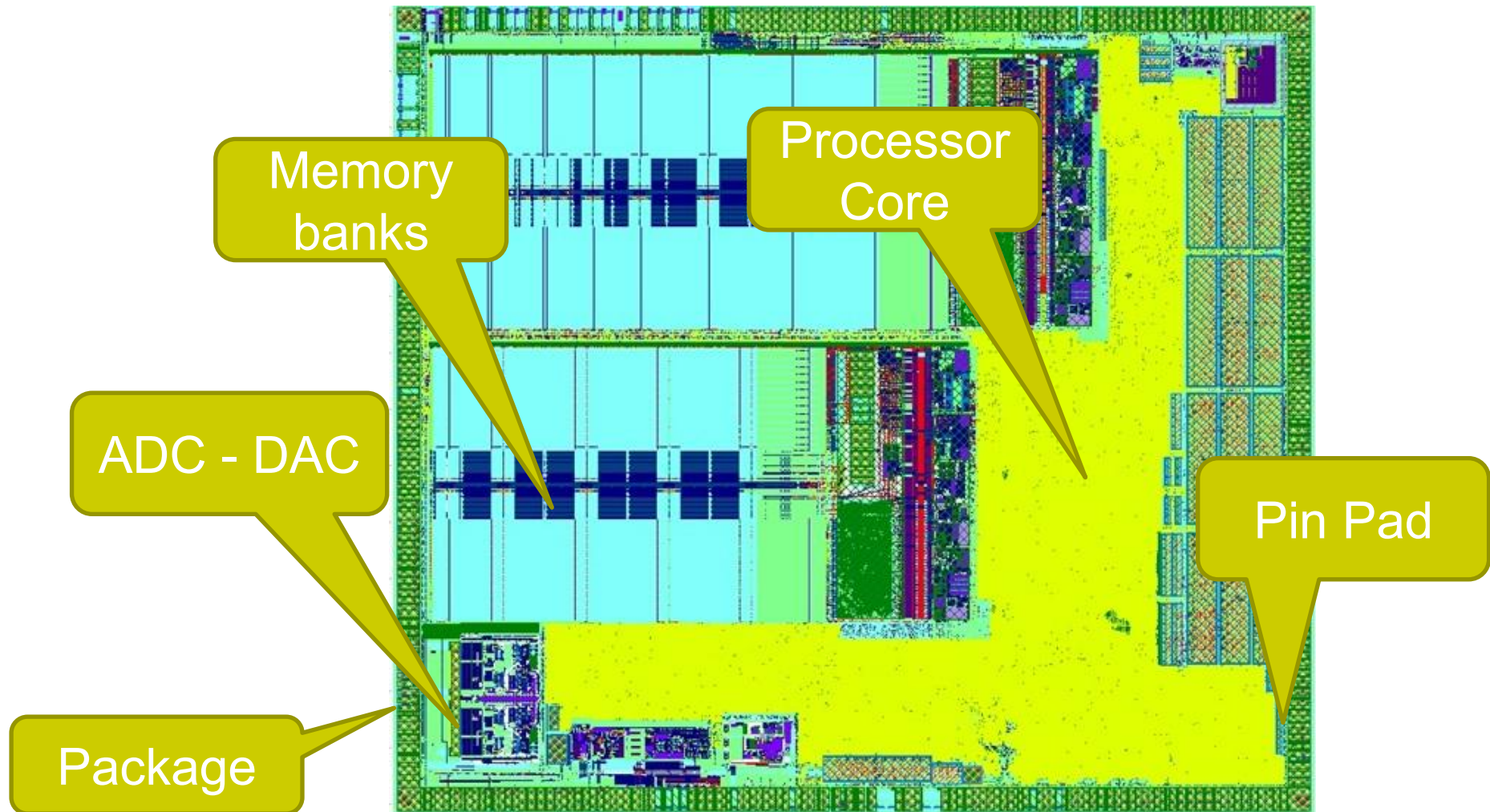


System on Chip

- An SoC is an entire system integrated in a single piece of silicon.



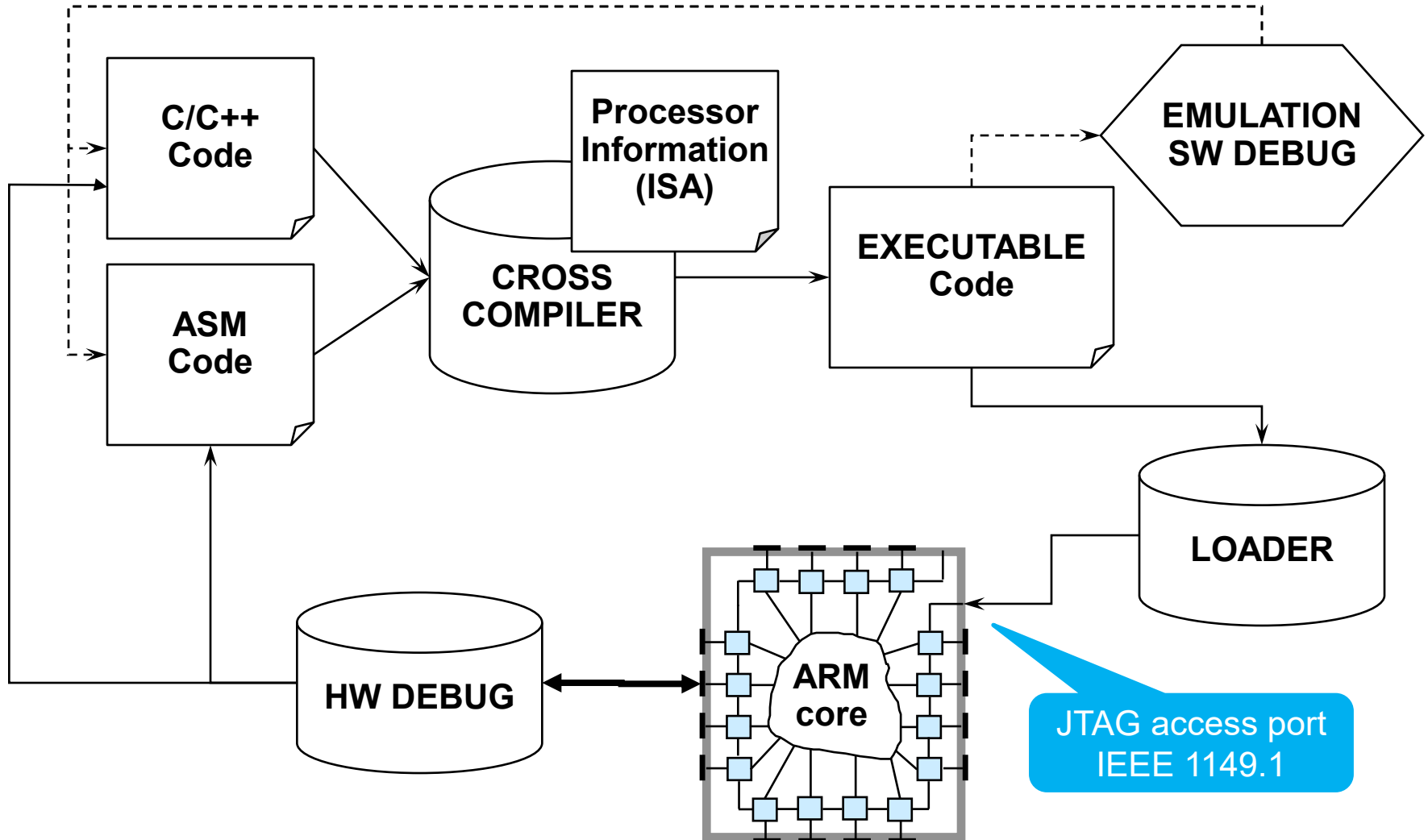
An example of SoC layout



ARM-based commercial SoCs

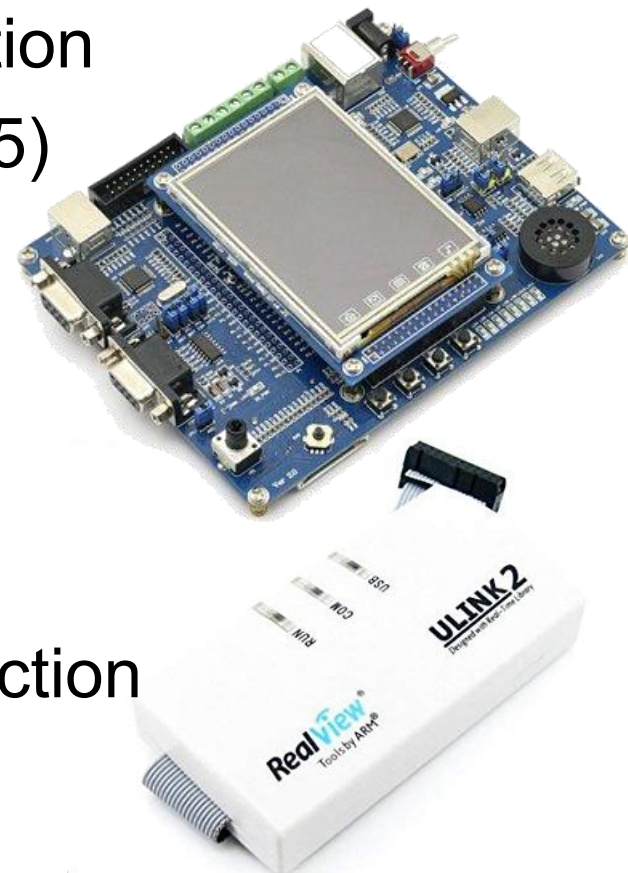
- NXP LPC176x: https://www.nxp.com/docs/en/data-sheet/LPC1769_68_67_66_65_64_63.pdf (page 6)
- ST Microelectronics STR75xFxx: <https://www.st.com/resource/en/datasheet/str750fv0.pdf> (page 11)
- Qualcomm Snapdragon 600 APQ8064 https://www.qualcomm.com/content/dam/qcomm-martech/dm-assets/documents/snapdragon_600_apq_8064_data_sheet.pdf (page 17)
- ...and many others...

Toolchain



Case of study

- Cross compilation + SW debug: KEIL uVision
 - <http://www.keil.com/> MDK-ARM
 - trial version with 32K code limitation
 - latest version 5.43a (August 2025)
- Development board: Landtiger
 - Based on NXP LPC1768 SoC
 - ARM Cortex-M3 core
- Hardware debug: RealView
 - with ULINK2 JTAG-based connection



Suggested links

- The following links describe the instruction set and give useful information about:
 - ARM compiler
<https://developer.arm.com/documentation/dui0473/m/preface>
 - Cortex-M3
<https://developer.arm.com/documentation/ddi0337/e/Introduction/Instruction-set-summary?lang=en>
- The links will be available during the exam.